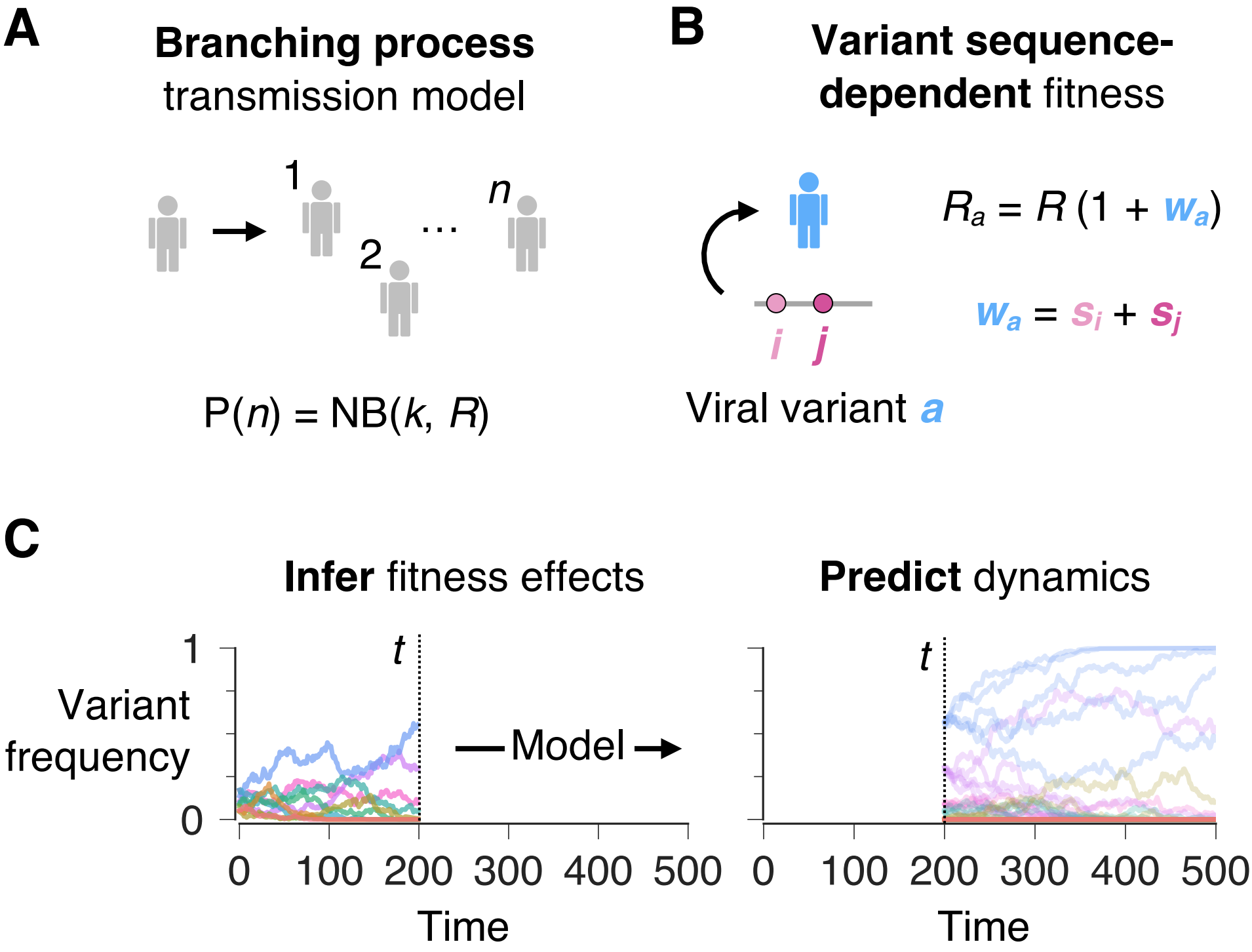

A
Branching process
transmission model
1
2
...
n
P(n) = NB(k, R)
B
Variant sequence-dependent fitness
$R_a = R\,(1 + w_a)$
$w_a = s_i + s_j$
i
j
Viral variant a
C
Infer fitness effects
Predict dynamics
Variant frequency
1
0
t
Model
0
100
200
300
400
500
Time